

CHANGCHUN, China

WMO: 541610

Lat: 43.901N

Lon: 125.209E

Elev: 781

StdP: 14.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.4	(c) -9.7	(d) -23.4	(e) 1.6	(f) -10.7	(g) -19.5	(h) 1.9	(i) -6.3	(j) 21.6	(k) 19.3	(l) 18.9	(m) 19.4	(n) 4.5	(o) 220	(p) 0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	88.2	69.6	86.0	69.8	83.9	69.2	76.0	82.9	74.4	81.1	72.9	79.4	10.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	129.4	80.3	72.0	121.9	78.6	70.7	116.4	77.4	40.2	82.9	38.6	81.5	37.3	79.4	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	19.3	16.6	-19.5	92.8	6.5	2.8	-24.1	94.8	-27.9	96.4	-31.6	98.0	-36.3	100.0
			-19.8	77.7	6.4	2.1	-24.4	79.3	-28.2	80.5	-31.7	81.7	-36.4	83.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	5.3	14.2	29.4	47.4	60.7	70.5	74.8	72.0	61.5	46.1	26.4	10.0		
	DBStd	25.74	8.87	10.48	10.32	9.39	7.33	5.87	4.18	4.78	6.59	8.99	11.44	9.74		
	HDD50	5328	1385	1001	642	158	5	0	0	0	4	183	711	1239		
	HDD65	8698	1850	1421	1105	532	176	16	0	7	142	587	1158	1704		
	CDD50	2899	0	0	2	80	337	616	768	682	350	62	2	0		
	CDD65	794	0	0	0	3	42	183	304	223	38	1	0	0		
	CDH74	6069	0	0	0	43	453	1533	2278	1488	262	12	0	0		
	CDH80	1600	0	0	0	4	110	491	632	335	28	0	0	0		
(14) Wind	WSAvg	7.5	6.4	7.5	8.6	9.8	9.1	7.3	6.5	5.8	6.3	7.5	7.8	6.9		
(15) Precipitation	PrecAvg	22.8	0.1	0.2	0.4	0.9	1.7	3.3	7.4	5.0	2.1	1.1	0.4	0.2		
	PrecMax	32.4	0.4	0.7	1.3	3.0	3.5	7.8	17.5	13.8	6.2	4.1	1.3	0.6		
	PrecMin	13.0	0.0	0.0	0.0	0.0	0.2	0.7	1.7	0.9	0.5	0.0	0.0	0.0		
	PrecStd	4.2	0.1	0.2	0.4	0.7	0.9	1.6	3.5	3.1	1.3	0.9	0.3	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	45.6	63.2	79.4	87.6	93.2	91.4	88.4	83.0	75.4	60.4	41.1		
		MCWB	28.6	35.3	46.5	54.9	61.2	66.3	73.7	73.5	66.8	58.4	49.3	37.6		
	2%	DB	27.9	39.8	55.3	73.3	82.9	88.8	88.0	86.0	80.2	69.5	53.4	33.7		
		MCWB	23.8	32.1	41.5	52.7	59.1	65.0	73.1	73.0	65.3	55.2	44.5	29.7		
	5%	DB	23.8	35.8	50.0	68.4	79.2	85.7	85.9	83.9	77.0	64.6	48.4	29.8		
		MCWB	20.4	29.2	38.0	50.2	57.8	65.1	72.5	72.1	62.9	52.5	41.6	26.4		
	10%	DB	19.7	31.7	45.6	64.2	75.1	82.3	83.7	81.4	73.9	60.9	43.1	25.2		
		MCWB	16.8	26.5	35.7	47.7	56.1	65.0	72.0	70.6	61.2	50.8	37.3	22.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.8	36.6	47.7	58.9	65.4	73.0	78.8	78.3	71.2	61.0	50.6	36.9		
		MCDB	32.5	43.0	61.1	72.2	79.5	80.8	86.3	83.9	78.1	69.1	57.7	40.6		
	2%	WB	24.1	32.7	42.8	54.5	62.0	70.4	76.7	75.8	68.4	57.5	45.4	30.2		
		MCDB	27.5	38.9	54.0	70.2	76.9	80.9	83.7	82.7	76.0	66.2	52.1	33.0		
	5%	WB	20.4	29.6	39.1	51.3	60.1	68.8	75.0	74.3	66.1	54.6	41.8	26.2		
		MCDB	23.7	35.6	48.7	66.6	73.4	79.0	81.8	80.8	73.3	63.1	47.5	29.4		
	10%	WB	16.9	26.1	36.0	48.6	58.0	67.3	73.6	72.8	63.7	51.7	37.7	22.1		
		MCDB	19.9	31.7	45.0	62.0	71.9	77.0	80.5	78.6	71.6	59.7	43.3	25.2		
(35) Mean Daily Temperature Range	MDBR		18.0	19.5	19.1	21.0	20.2	18.1	14.9	15.8	20.5	19.9	16.7	16.9		
	5% DB	MCDBR	21.2	21.4	24.1	26.4	25.3	22.7	17.9	18.3	22.6	23.9	20.4	19.9		
		MCWBR	18.3	15.9	14.7	13.8	10.7	8.6	7.7	8.4	10.7	13.3	14.4	17.5		
	5% WB	MCDBR	20.9	20.9	22.5	23.4	21.0	16.8	14.6	14.4	18.7	21.2	19.4	19.8		
(40) Clear-Sky Solar Irradiance	taub		0.328	0.358	0.436	0.532	0.599	0.660	0.623	0.546	0.474	0.492	0.399	0.343		
	taud		2.072	2.021	1.897	1.749	1.664	1.629	1.741	1.902	2.001	1.845	1.990	2.037		
	Ebn at Noon		240	255	248	234	223	210	216	228	235	203	207	218		
	Edh at Noon		33	42	54	68	76	80	70	58	49	50	36	31		
(44) All-Sky Solar Radiation	RadAvg		705	1030	1367	1609	1800	1747	1618	1536	1396	1001	672	576		
	RadStd		60	71	68	116	136	163	148	120	100	67	59	46		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-2.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page